Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 13: Equivalent Resistances and Linearity

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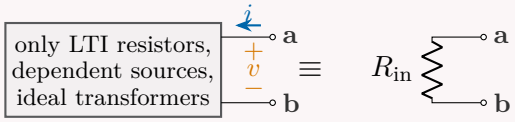
Colour code used in these notes and in the slides			
current i	voltage v, node voltage e	a Δ	a Y

Motivation. Previously, we simplified circuits with equivalent one-ports: series and parallel connections and source transformation. Two cases need more. A one-port with dependent sources is not a series–parallel connection of resistors. A bridge has no two resistors in series or in parallel. The first part finds one equivalent resistor in both cases: the **input resistance** of a one-port, and the **Δ–Y transform** of three resistors. The second part shows that an LTI resistive circuit is **linear** in its independent sources. Then **superposition** solves a circuit with several sources one source at a time.

1. The input resistance of a one-port

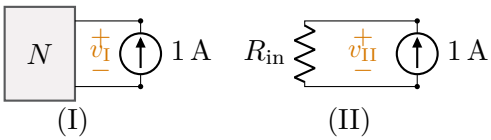
Input resistance

Let a one-port N contain only LTI resistors, LTI dependent sources and ideal transformers, with every control variable inside N. Then N is equivalent to a single LTI resistor R_{in}, the **input resistance** of N: $v = R_{\text{in}}\, i$ at its port.

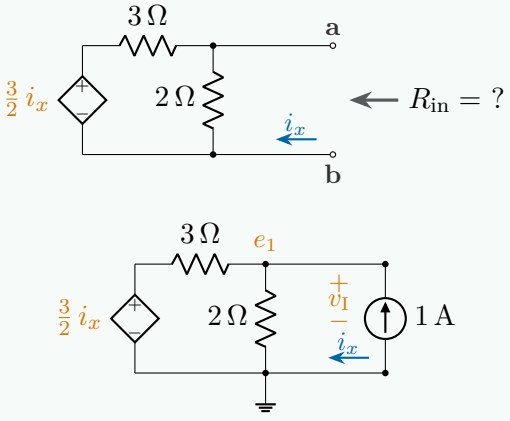


With dependent sources, series and parallel rules are not enough to find R_{in}. A **test source** finds it. Connect a 1 A current source to the port of N (circuit I) and to R_{in} (circuit II). Since $N \equiv R_{\text{in}}$, the port voltages are equal: $v_{\text{I}} = v_{\text{II}} = R_{\text{in}} \cdot 1\,\text{A}$, so

$$R_{\text{in}} = \frac{v_{\text{I}}}{1\,\text{A}}: \quad \text{solve circuit (I).}$$



Example 13.1 — an input resistance



Question. Find R_{in}.

Answer. Connect a 1 A test source and take the bottom node as the ground. KCL at node 1:

$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0.$$

The test current enters at a and leaves at b, so $i_x = -1\,\text{A}$. First plug it in:

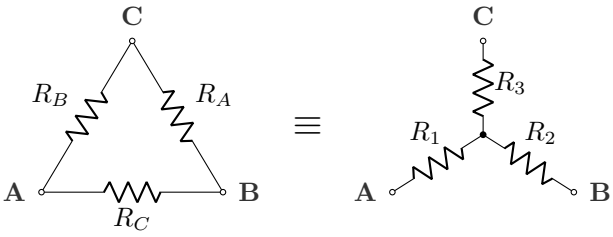
$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; \frac{5}{6}\, e_1 = \frac{1}{2} \;\Rightarrow\; e_1 = \frac{3}{5}\,\text{V}.$$

So $v_{\text{I}} = e_1 = \frac{3}{5}\,\text{V}$ and $R_{\text{in}} = \frac{3}{5}\,\Omega$.

With the dependent source replaced by a short circuit, R_{in} would be $3 \parallel 2 = \frac{6}{5}\,\Omega$: the dependent source changes the input resistance.

2. The Δ–Y transform

Three resistors may form a Δ or a Y between three terminals A, B, C. In the Δ, R_A is opposite A (between B and C), R_B is opposite B and R_C is opposite C. In the Y, R_1, R_2, R_3 are at A, B, C.



Δ to Y. Given R_A, R_B, R_C, find R_1, R_2, R_3. If the two are equivalent, the resistance between any two terminals, with the third one open, is the same. Between A and B, the Δ is R_C in parallel with $R_A + R_B$, and the Y is $R_1 + R_2$:

$$\frac{R_C(R_A+R_B)}{R_A+R_B+R_C} = R_1+R_2 \quad (1), \qquad \frac{R_A(R_B+R_C)}{R_A+R_B+R_C} = R_2+R_3 \quad (2),$$

$$\frac{R_B(R_C+R_A)}{R_A+R_B+R_C} = R_3+R_1 \quad (3).$$

Half of the sum of (1), (2) and (3) gives

$$R_1+R_2+R_3 = \frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \qquad (4),$$

and (4)−(2), (4)−(3), (4)−(1) give the result.

Δ to Y

$$R_1 = \frac{R_BR_C}{R_A+R_B+R_C}, \qquad R_2 = \frac{R_CR_A}{R_A+R_B+R_C}, \qquad R_3 = \frac{R_AR_B}{R_A+R_B+R_C}$$

The Y resistor at a terminal is the product of the two Δ resistors at that terminal, divided by the sum of the Δ resistors.

Y to Δ. Given R_1, R_2, R_3, find R_A, R_B, R_C. From the result above,

$$S = R_1R_2+R_2R_3+R_3R_1 = \frac{R_AR_BR_C\,(R_A+R_B+R_C)}{(R_A+R_B+R_C)^2} = \frac{R_AR_BR_C}{R_A+R_B+R_C},$$

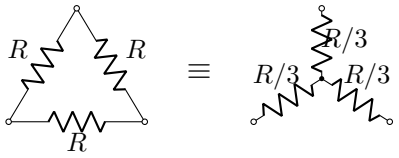
$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A+R_B+R_C} = S.$$

Y to Δ

$$R_A = \frac{S}{R_1}, \qquad R_B = \frac{S}{R_2}, \qquad R_C = \frac{S}{R_3}, \qquad S = R_1R_2+R_2R_3+R_3R_1$$

A Δ resistor is the sum of the products of the Y resistors, divided by the Y resistor opposite to it.

Equal resistors. A Δ of three equal resistors R is equivalent to a Y of three resistors $R \cdot R/(3R) = R/3$. Conversely, a Y of three resistors R is equivalent to a Δ of three resistors $3R$.

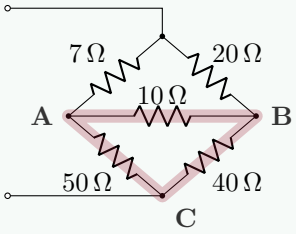


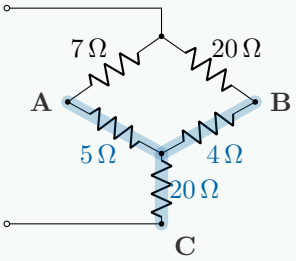
Exercise. Write both transforms with conductances, $G_\square = 1/R_\square$. *Answer:* Y to Δ becomes $G_A = G_2G_3/(G_1+G_2+G_3)$, and so on; it has the form of Δ to Y with resistances.

3. An example: a bridge

Example 13.2 — a bridge

Question. Find the resistance R between the two terminals. No two resistors are in series or in parallel.





Answer. Transform the Δ A–B–C into a Y: $R_C = 10\,\Omega$, $R_B = 50\,\Omega$, $R_A = 40\,\Omega$, and $R_A + R_B + R_C = 100\,\Omega$. So

$$R_1 = \frac{50 \cdot 10}{100} = 5\,\Omega, \qquad R_2 = \frac{10 \cdot 40}{100} = 4\,\Omega, \qquad R_3 = \frac{40 \cdot 50}{100} = 20\,\Omega.$$

In the new circuit, $7 + 5 = 12\,\Omega$ and $20 + 4 = 24\,\Omega$ are in parallel, $12 \parallel 24 = 8\,\Omega$, and this is in series with $20\,\Omega$:

$$R = 8 + 20 = 28\,\Omega.$$

Check. With node voltages, a 1 A test source between the terminals gives a terminal voltage of 28 V. ✓

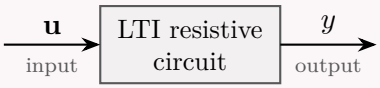
4. Linearity

In an LTI resistive circuit, the independent sources are the **inputs**. Collect the voltages of the independent voltage sources and the currents of the independent current sources in the **input vector** $\mathbf{u} = [\,u_1 \;\cdots\; u_m\,]^T$. An **output** y is any branch voltage or current.

Linearity

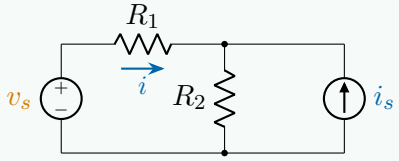
For each output y of an LTI resistive circuit, there is a row vector $\mathbf{k} = [\,k_1 \;\cdots\; k_m\,]$ such that

$$y = \mathbf{k}\,\mathbf{u} = k_1 u_1 + k_2 u_2 + \cdots + k_m u_m.$$



y is a function of $\mathbf{u}$: $y = y(\mathbf{u})$

Example 13.3 — the row vector k



Question. The inputs are $\mathbf{u} = [\,v_s \;\; i_s\,]^T$. For the output $y = i$, find k_1, k_2 with $i = k_1 v_s + k_2 i_s$.

Answer. KVL around the left mesh, with $i + i_s$ down through R_2:

$$-v_s + R_1 i + R_2(i + i_s) = 0,$$

$$i = \frac{1}{R_1 + R_2}\, v_s - \frac{R_2}{R_1 + R_2}\, i_s,$$

so $k_1 = 1/(R_1 + R_2)$ and $k_2 = -R_2/(R_1 + R_2)$.

The implication of linearity

For input vectors $\mathbf{u}$, $\mathbf{w}$ and numbers α, β:

$$y(\alpha\,\mathbf{u} + \beta\,\mathbf{w}) = \alpha\,y(\mathbf{u}) + \beta\,y(\mathbf{w}).$$

5. Superposition

Split the input vector into one source at a time. With two sources,

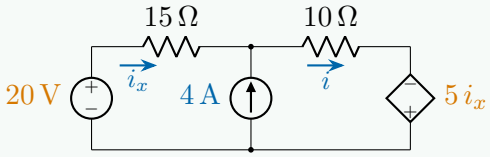
$$y\left(\begin{bmatrix} v_s \\ i_s \end{bmatrix}\right) = y\left(\begin{bmatrix} v_s \\ 0 \end{bmatrix}\right) + y\left(\begin{bmatrix} 0 \\ i_s \end{bmatrix}\right):$$

the output is the sum of the outputs with one independent source at a time. The other sources are **killed**.

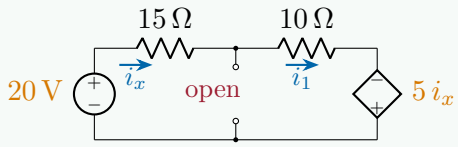
Killing a source

Killing an independent voltage source ($v_s = 0$) replaces it with a **short circuit**. Killing an independent current source ($i_s = 0$) replaces it with an **open circuit**. Do **not** kill dependent sources: they are not inputs.

Example 13.4 — superposition



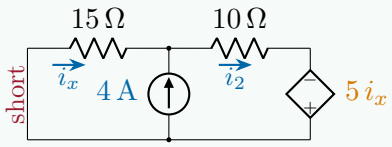
Question. Find i by superposition.
Answer. $i = i_1 + i_2$: i_1 with the 4 A source killed, i_2 with the 20 V source killed.



Kill the current source (open). One loop, $i_x = i_1$:

$$-20 + 15\,i_x + 10\,i_1 - 5\,i_x = 0$$

$$\Rightarrow\ 20\,i_1 = 20\ \Rightarrow\ i_1 = 1\,\text{A}.$$



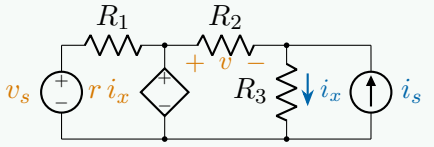
Kill the voltage source (short). KCL: $i_x = i_2 - 4$:

$$15\,i_x + 10\,i_2 - 5\,i_x = 0$$

$$\Rightarrow\ 20\,i_2 = 40\ \Rightarrow\ i_2 = 2\,\text{A}.$$

So $i = i_1 + i_2 = 3\,\text{A}$. The dependent source $5\,i_x$ stays in both circuits.

Example 13.5 — three experiments



	v_s	i_s	v
Exp. 1	12 V	0	v_1
Exp. 2	−6 V	6 A	v_2
Exp. 3	4 V	4 A	v_3

Question. Three experiments are made on the circuit. Express v_3 in terms of v_1 and v_2.
Answer. By linearity, $v = \mathbf{k}\,[\,v_s \;\; i_s\,]^T$ for some $\mathbf{k}$. Write the third input with the first two:

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \ \Rightarrow\ 6\beta = 4,\quad 12\alpha - 6\beta = 4\ \Rightarrow\ \alpha = \beta = \tfrac{2}{3}.$$

Then $v_3 = \alpha\,v_1 + \beta\,v_2 = \frac{2}{3}\,v_1 + \frac{2}{3}\,v_2$. Equivalently, $\mathbf{k} = [\,v_1 \;\; v_2\,]\,\left[\begin{smallmatrix} 12 & -6 \\ 0 & 6 \end{smallmatrix}\right]^{-1}$. No element value is needed.

Summary

In the circuit	**What to do**
a one-port of resistors and dependent sources	connect a 1 A test source; $R_{\text{in}} = v/1\,\text{A}$
a Δ	Y: $R_1 = R_B R_C/(R_A + R_B + R_C)$, $R_2 = R_C R_A/(R_A + R_B + R_C)$, $R_3 = R_A R_B/(R_A + R_B + R_C)$
a Y	Δ: $R_A = (R_1 R_2 + R_2 R_3 + R_3 R_1)/R_1$, and so on
equal resistors	Δ with $R \equiv$ Y with $R/3$
a linear circuit	$y = \mathbf{k}\,\mathbf{u}$; $y(\alpha\mathbf{u} + \beta\mathbf{w}) = \alpha\, y(\mathbf{u}) + \beta\, y(\mathbf{w})$
several independent sources	superposition: one source at a time; a killed voltage source is a short circuit, a killed current source an open circuit; dependent sources stay